DD_SUB-WF-ACTIVATE-01-FLOW-WIK - WIK Activation BPMN

Developer implementation guide: `DD_SUB-WF-ACTIVATE-01-FLOW-WIK-DEV_IMPL_GUIDE-v1.0.md`.

1. Purpose

This DD defines the WIK operator BPMN for subscription activation.

Process key:

```
sub-activate-wik
```

This BPMN implements the parent activation contract from `DD_SUB-WF-ACTIVATE-01-Subscription_Activation-v1.0.md` for WIK.

The WIK activation flow is pay-first:

```
validate -> PENDING_ACTIVATION -> billing pre-authorize -> wait for payment -> confirm charge -> network activation -> Hom
```

2. Ownership

This DD owns:

- WIK-specific activation BPMN process key.
- WIK happy-path BPMN sequence.
- WIK payment wait and timeout behavior.
- WIK compensation paths.
- WIK worker-topic composition.
- WIK message catches.
- WIK use of `rules.subscription.activate.wik`.
- WIK activation examples.

This DD does not own:

- Activation API contract and config table. See parent activation DD.
- Shared operation framework. See `SUB-WF-FRAMEWORK-01`.
- Worker contracts. See `DD_SUB-WF-FRAMEWORK-01-WORKERS`.
- Worker implementation code.
- SUB-LM, BIL, FUL, OSR, NOT internals.

3. Required Runtime Configuration

3.1 Framework Operation Config

Required row:

```
INSERT INTO subscription_operation_config (
  operator_code,
  operation_kind,
  default_bpmn_process_key,
  operation_timeout_seconds,
  billing_call_timeout_seconds,
  fulfillment_call_timeout_seconds,
  feature_flags,
  enabled,
  updated_by
)
VALUES (
  'WIK',
  'ACTIVATE',
  'sub-activate-wik',
  900,
```

```
  60,
  120,
  '{"pay_first_required": true, "payment_timeout_days": 7}',
  TRUE,
  'migration'
);
```

Meaning:

- WIK activation starts `sub-activate-wik`.
- WIK requires pay-first activation.
- Customer has 7 days to complete payment before the BPMN compensates.

3.2 Activation Config

Required row:

```
INSERT INTO subscription_activation_config (
  operator_code,
  cycle_anchor_overflow_default,
  customer_self_service_enabled,
  sales_field_activation_enabled,
  scheduled_activation_enabled,
  equipment_binding_validation_strict,
  backdating_allowed,
  backdating_max_days,
  backdating_required_role,
  updated_by
)
VALUES (
  'WIK',
  'PRORATE_PARTIAL_CYCLE',
  FALSE,
  TRUE,
  TRUE,
  TRUE,
  TRUE,
  7,
  'SUBSCRIPTION_ADMIN',
  'migration'
);
```

3.3 Drools Package

Required package:

```
rules.subscription.activate.wik
```

Required decisions:

```
validate-preconditions
validate-backdating
resolve-cycle-anchor-overflow
validate-equipment-bindings
```

4. BPMN Start Variables

The framework starts Camunda with these variables.

Variable	Purpose	Example
`operationId`	Links BPMN to `subscription_operation`.	`subop_01HZ_KAMAU_ACT`
`operatorCode`	Operator scope.	`WIK`
`subscriptionId`	Subscription being activated.	`sub_01HZ_KAMAU_FIBER`
`operationKind`	Always `ACTIVATE`.	`ACTIVATE`
`customerId`	Customer reference.	`cust_01HZ_KAMAU`
`homepassId`	HomePass to activate at.	`hp_01HZ_KAREN_017`
`packageRef`	Package being activated.	`pkg_INET_VOIP_RES`
`packageVersionId`	Package version pinned at activation.	`pkgver_003`
`billingMode`	Billing mode on subscription.	`PREPAID`
`activationTrigger`	Trigger channel.	`SALES_FIELD`
`intendedActivationDate`	Requested business activation date.	`2026-05-18`

Variable	Purpose	Example
`resolvedActivatedAt`	Activation timestamp resolved by parent validation.	`2026-05-18T10:00:00Z`
`resolvedCycleAnchorDay`	SUB-LM-safe cycle anchor.	`18`
`cycleAnchorOverflowApplied`	Overflow policy applied, if any.	`null`
`partialCycleDays`	Partial days for billing context.	`0`
`equipmentBindings`	Equipment to bind.	`[{"equipmentRequirement":"GPON_ONT","equipmentRef":"ont_01HZ"}]`
`initiatingActorUserId`	Actor user/service.	`sales-agent-wik-007`
`initiatingActorRole`	Actor role.	`SALES_AGENT`
`initiatingWorkflowKind`	Upstream workflow.	`FUL_02_OBR_01`
`initiatingWorkflowRef`	Upstream reference.	`obr_01HZ_KAMAU`

5. Worker Topics Used

These are logical Camunda external-task topics, not separate deployments.

Worker topic	Why WIK activation uses it
`drools-decide`	Re-check WIK activation policy and route based on decisions.
`sub-lm-put-pending-status`	Move subscription to `PENDING_ACTIVATION`.
`sub-lm-compute-cycle-window`	Compute cycle window for billing context.
`bil01-charge-pre-authorize`	Create activation/installation/first-cycle pay-first charge.
`bil01-confirm-charge`	Confirm charge after payment message arrives.
`ful-call-activate`	Ask FUL-03 to provision WIK service.
`ful-call-claim-homepass`	Claim HomePass active termination point through FUL-07.
`osr-equipment-bind`	Bind ONT/ATA/STB equipment to subscription.
`sub-lm-put-master-fields`	Store activation master fields.
`sub-lm-commit-state`	Commit subscription to `ACTIVE`.
`kafka-emit`	Write activation success/failure event to outbox.
`notification-send`	Send welcome/failure notification without failing BPMN on notification failure.
`subscription-operation-finalize`	Set operation terminal state.
`bil01-revert-pending-charge`	Compensation for unpaid/failed activation charge.
`sub-lm-revert-to-prior`	Compensation to return subscription to `CREATED`.
`osr-equipment-unbind`	Compensation if equipment was bound and later needs rollback.

6. Happy Path BPMN

```
Start
  -> update operation state VALIDATING
  -> drools-decide
  -> gateway: eligible?
      no -> activation rejected path
      yes
  -> update operation state PENDING_STATE_FLIP
  -> sub-lm-put-pending-status
  -> sub-lm-compute-cycle-window
  -> update operation state BILLING_CALL
  -> bil01-charge-pre-authorize
  -> update operation state AWAITING_PAYMENT
  -> wait for payment-confirmed message
  -> bil01-confirm-charge
  -> update operation state FULFILLMENT_CALL
  -> ful-call-activate
  -> update operation state AWAITING_FULFILLMENT_RESPONSE
  -> wait for fulfillment-completed message
  -> ful-call-claim-homepass
  -> osr-equipment-bind
  -> update operation state COMMITTING_FINAL_STATE
  -> sub-lm-put-master-fields
  -> sub-lm-commit-state
  -> update operation state EMITTING_EVENT
  -> kafka-emit SubscriptionActivated
```

```
-> notification-send
-> subscription-operation-finalize COMPLETED
-> End
```

7. Step Details

7.1 `drools-decide`

Inputs:

```
{
  "kjarPackage": "rules.subscription.activate.wik",
  "requestedDecisions": [
    "validate-preconditions",
    "validate-backdating",
    "resolve-cycle-anchor-overflow",
    "validate-equipment-bindings"
  ],
  "factsJson": "{...activation facts...}"
}
```

Expected output:

```
{
  "decisionsJson": {
    "eligible": true,
    "resolvedActivatedAt": "2026-05-18T10:00:00Z",
    "resolvedCycleAnchorDay": 18,
    "partialCycleDays": 0,
    "equipmentValid": true
  }
}
```

If `eligible = false`, route to rejection path.

7.2 `sub-lm-put-pending-status`

Inputs:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "toStatus": "PENDING_ACTIVATION",
  "currentTransitionType": "ACTIVATE",
  "currentTransitionReasonCode": "SALES_FIELD"
}
```

Output consumed later:

```
{
  "priorStatusSnapshot": {
    "statusCode": "CREATED"
  }
}
```

7.3 `bil01-charge-pre-authorize`

Inputs:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "chargeCode": "ACTIVATION_INTENT",
  "currency": "KES",
  "chargeContext": {
    "activationTrigger": "SALES_FIELD",
    "resolvedActivatedAt": "2026-05-18T10:00:00Z",
    "partialCycleDays": 0,
    "equipmentBindings": [
      {
        "equipmentRequirement": "GPON_ONT",
        "equipmentRef": "ont_01HZ_KAMAU"
      }
    ]
  },
  "cycleWindow": {
    "cycleStart": "2026-05-18",
    "cycleEnd": "2026-06-17"
  }
}
```

Expected output:

```json
{
  "billableEventId": "bevt_01HZ_KAMAU_ACT",
  "customerChargeAmount": 7000.00,
  "currency": "KES"
}
```

7.4 Payment Message Catch

Message name:

```
payment-confirmed
```

Correlation key:

```
billableEventId
```

Timer:

```
P7D
```

If timer fires, route to payment-timeout compensation.

7.5 `ful-call-activate`

Inputs:

```json
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "activationCorrelationId": "subop_01HZ_KAMAU_ACT:activate",
  "equipmentBindings": [
    {
      "equipmentRequirement": "GPON_ONT",
      "equipmentRef": "ont_01HZ_KAMAU"
    }
  ],
  "policy": {
    "operatorCode": "WIK",
    "technology": "GPON"
  }
}
```

Expected output:

```json
{
  "activationFulfillmentId": "ful_01HZ_KAMAU_ACT",
  "activationTerminalState": "SUBMITTED"
}
```

The BPMN then waits for `fulfillment-completed` or `fulfillment-failed`.

7.6 `sub-lm-put-master-fields`

Inputs:

```json
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "fieldsJson": {
    "activated_at": "2026-05-18T10:02:22Z",
    "cycle_anchor_day": 18,
    "package_version_id": "pkgver_003",
    "equipment_bindings": [
      {
        "equipmentRequirement": "GPON_ONT",
        "equipmentRef": "ont_01HZ_KAMAU"
      }
    ],
    "contract_started_at": "2026-05-18T10:02:22Z",
    "contract_end_at": "2027-05-18T10:02:22Z"
  }
}
```

7.7 `kafka-emit`

Success topic:

```
sophix.subscription.subscription.activated
```

Payload is `SubscriptionActivated` from the parent activation DD.

8. Message Catches

Message	Sent by	Correlation key	BPMN route
`payment-confirmed`	BIL/payment correlator	`billableEventId`	Continue to `bil01-confirm-charge`.
`fulfillment-completed`	FUL-03 correlator	`activationFulfillmentId`	Continue to HomePass/equipment/final commit.
`fulfillment-failed`	FUL-03 correlator	`activationFulfillmentId`	Route to fulfillment-failed compensation.
`external-cancel-requested`	Framework cancel API	`operationId`	Route to cancellation compensation.

9. Compensation Paths

9.1 Rejection Before Pending State

When validation rejects before `sub-lm-put-pending-status`, execute:

Order	Worker topic	Purpose
1	`kafka-emit`	Emit `SubscriptionActivationFailed`.
2	`subscription-operation-finalize`	Finalize operation as `FAILED`.

No SUB-LM revert is needed because subscription is still `CREATED`.

9.2 Payment Timeout

When `payment-confirmed` does not arrive within `P7D`, execute:

Order	Worker topic	Purpose
1	`bil01-revert-pending-charge`	Cancel unpaid activation charge.
2	`sub-lm-revert-to-prior`	Return subscription to `CREATED`.
3	`kafka-emit`	Emit `SubscriptionActivationFailed`.
4	`subscription-operation-finalize`	Finalize operation as `FAILED`.

Failure code:

```
ACTIVATION_PAYMENT_TIMEOUT
```

9.3 Fulfillment Failed

When FUL-03 sends `fulfillment-failed`, execute:

Order	Worker topic	Purpose
1	`bil01-revert-pending-charge`	Cancel/reverse pending activation charge.
2	`sub-lm-revert-to-prior`	Return subscription to `CREATED`.
3	`kafka-emit`	Emit `SubscriptionActivationFailed`.
4	`subscription-operation-finalize`	Finalize operation as `FAILED`.

Failure code:

```
FULFILLMENT_FAILED
```

9.4 Equipment Bind Failed

When OSR equipment binding fails after fulfillment succeeded, execute:

Order	Worker topic	Purpose
1	`ful-call-terminate` or approved rollback topic	Roll back network activation if supported.
2	`bil01-revert-pending-charge`	Cancel/reverse activation charge according to BIL rules.
3	`sub-lm-revert-to-prior`	Return subscription to `CREATED`.
4	`kafka-emit`	Emit `SubscriptionActivationFailed`.
5	`subscription-operation-finalize`	Finalize operation as `FAILED`.

Failure code:

```
EQUIPMENT_BINDING_INVALID
```

9.5 Notification Failed

Notification failure does not fail activation.

Behavior:

- Log failure.
- Let NOT-01 retry according to its own rules.
- Continue to operation finalization.

10. BPMN XML Fragments

10.1 Pay-First Gate

```
<bpmn:serviceTask id="Task_BilPreAuthorize"
                  name="Pre-authorize activation charge"
                  camunda:type="external"
                  camunda:topic="bil01-charge-pre-authorize" />

<bpmn:intermediateCatchEvent id="Catch_PaymentConfirmed"
                             name="Payment confirmed">
  <bpmn:messageEventDefinition messageRef="Message_payment_confirmed" />
</bpmn:intermediateCatchEvent>

<bpmn:boundaryEvent id="Boundary_PaymentTimeout"
                    attachedToRef="Catch_PaymentConfirmed">
  <bpmn:timerEventDefinition>
    <bpmn:timeDuration>P7D</bpmn:timeDuration>
  </bpmn:timerEventDefinition>
</bpmn:boundaryEvent>
```

10.2 Fulfillment Wait

```
<bpmn:serviceTask id="Task_FulActivate"
                  name="Activate service"
                  camunda:type="external"
                  camunda:topic="ful-call-activate" />

<bpmn:intermediateCatchEvent id="Catch_FulfillmentCompleted"
                             name="Fulfillment completed">
  <bpmn:messageEventDefinition messageRef="Message_fulfillment_completed" />
</bpmn:intermediateCatchEvent>

<bpmn:intermediateCatchEvent id="Catch_FulfillmentFailed"
                             name="Fulfillment failed">
  <bpmn:messageEventDefinition messageRef="Message_fulfillment_failed" />
</bpmn:intermediateCatchEvent>
```

11. Worked Example

Kamau signs up for WIK fiber and voice.

Runtime sequence:

1. FUL-02 creates subscription `sub_01HZ_KAMAU_FIBER` in `CREATED`.
2. Field sales sends activation request.
3. Framework starts `sub-activate-wik`.
4. Drools accepts: subscription `CREATED`, package active, HomePass sellable, role `SALES_AGENT`.
5. SUB-LM moves subscription to `PENDING_ACTIVATION`.
6. BIL creates pay-first billable event for installation and first cycle.
7. Customer pays through M-Pesa.
8. Payment correlator sends `payment-confirmed`.
9. BIL confirms charge.

10. FUL-03 provisions GPON/voice service.
11. FUL correlator sends `fulfillment-completed`.
12. FUL-07 claims HomePass capacity.
13. OSR binds ONT/ATA equipment.
14. SUB-LM stores activation fields.
15. SUB-LM commits `ACTIVE`.
16. BPMN emits `SubscriptionActivated`.
17. NOT-01 sends welcome notification.
18. Framework finalizes operation as `COMPLETED`.

12. Monitoring

Monitor:

Metric	Why
WIK activation count	Business volume.
WIK activation failure count	Operational quality.
Payment timeout count	Customer/payment issue.
FUL-03 failure count	Network provisioning issue.
Average activation duration	SLA.
Operations stuck in `AWAITING_PAYMENT`	Payment/correlation issue.
Operations stuck in `AWAITING_FULFILLMENT_RESPONSE`	FUL/correlation issue.

13. Cross-DD Dependencies

DD/module	Relationship
Parent activation DD	Defines activation API, config, events, and generic rules.
Activation dev guide	Defines execution-style implementation flow.
SUB-WF framework	Starts and tracks operation.
WORKERS catalog	Defines worker topic contracts.
SUB-LM	Owns subscription state.
BIL-01	Activation billing and payment confirmation.
FUL-03	Network activation.
FUL-07	HomePass claim.
OSR-INSTANCE-01	Equipment bind.
NOT-01	Welcome notification.

14. Test Checklist

- [ ] BPMN deploys with process key `sub-activate-wik`.
- [ ] Happy path reaches `COMPLETED`.
- [ ] Payment timeout routes to compensation.
- [ ] Fulfillment failure routes to compensation.
- [ ] Equipment bind failure routes to compensation.
- [ ] Notification failure does not fail activation.
- [ ] `SubscriptionActivated` is emitted only after SUB-LM commits `ACTIVE`.
- [ ] Operation final state is set exactly once.